

AI-Based Programming Error Explanation System

Hello Everyone!

Presented by: **Yugavarshini R** · B.E. Electronics and Communication Engineering – ECE-C





Problem Statement

Beginner programmers struggle to understand error messages — they're **technical, confusing, and hard to interpret.**



Repeated Mistakes

Same errors, same confusion,
no progress



Dependency

Relying on teachers, friends, or
Google



Frustration

Learning stalls before it begins

Student → Error Message → Confusion → Repeated Mistakes

1. Empathize



Problems faced by beginner programmers:

- **Cryptic error messages**
- **Can't locate the error**
- **Don't know *why* it happened**
- **Wasted debugging time**
- **Dependency on others**

2. Define / Design

Beginner programmers need an intelligent system that can **identify errors**, explain them in **simple language**, and guide them toward a **solution**.

Design Objective:

1



Detect

2



Explain

3



Guide

4



Teach

3. Ideate

AI-Powered Programming Error Explanation System — Main Features:



Auto Error Detection



Error Location ID



Simple Explanation



Solution Suggestions



Interactive AI Assistant



Multi-Language Support

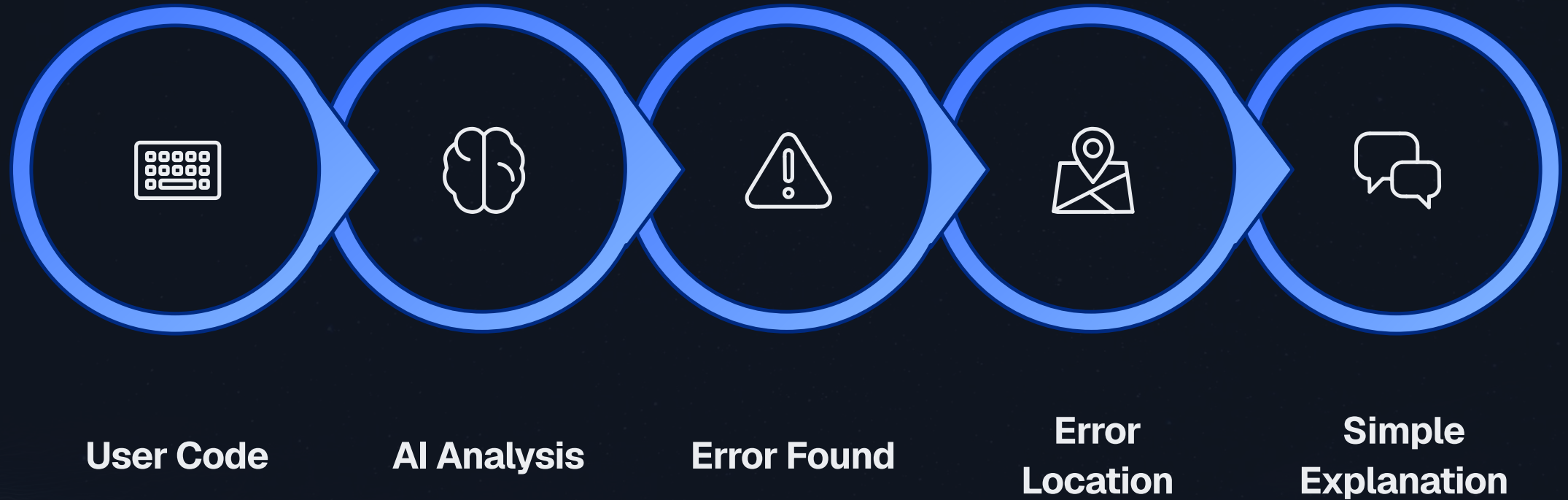


Learning Tips



Error History

System Workflow



From raw code to clear understanding — in seconds.

Technology & Innovation

Tech Stack

Frontend

HTML · CSS · JavaScript

Backend

Python / Flask

AI Engine

Machine Learning + NLP

Code Analysis

Parser / Compiler-based

Database

MySQL / SQLite

Innovation

Beginner-Friendly

Plain-language error explanations

Personalized Guidance

AI-generated, tailored to the student

Explains the *Why*

Not just what — but why it broke

Learn, Don't Copy

Builds understanding, not dependency

Multi-Language

Supports multiple programming languages

Conclusion

Design Thinking helped us understand the **real difficulties** faced by beginner programmers and develop a **user-centered AI solution**.

Don't just fix the error — understand the error.

Thank You!

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